Unified memory

The CPU and GPU tiles communicate through NVLink-C2C. NVIDIA specifies 600 GB/s of bidirectional interconnect bandwidth, while the LPDDR5X memory system provides up to 300 GB/s of peak memory bandwidth. These are different links in the architecture. The larger figure is not the GPU's memory bandwidth.

The platform supports up to 128 GB of unified memory, although this ASUS ProArt P16 test platform laptop has 64 GB. The GPU allocation was set to 32 GB. That is an allocation within the shared physical memory pool, not an additional 32 GB of dedicated VRAM. You can check and set the allocation via the NVIDIA App in a rather convenient manner. The benefit is straightforward. A workload that exceeds the graphics memory of a conventional laptop may fit into the GPU-accessible allocation here. A local language model also needs room for its context cache and runtime buffers, while Windows and other applications still need memory of their own.

Capacity and speed remain separate considerations. Dedicated laptop GPUs can have different memory bandwidth, power budgets and compute resources. Sharing memory also means that CPU and GPU activity can compete for resources. RTX Spark's architectural advantage is therefore most persuasive when a workload benefits from both the available capacity and NVIDIA's acceleration ecosystem.

Windows on Arm

RTX Spark runs native Arm applications and existing x86/x64 applications through Windows' Prism translation technology. That creates a broad route to software compatibility, but application architecture remains relevant to performance and reliability. A native ARM64 application avoids translating its CPU instructions. An x64 application uses Prism for that portion of execution. Supported GPU kernels still execute on the GPU, which is why a graphics-heavy application can perform differently from a CPU-heavy program under emulation.

ARM64EC adds another possibility, combining native Arm components with compatible x64 components. A native application label therefore does not necessarily establish that every plug-in or dependency follows the same path. NVIDIA and Microsoft have been working on scheduling, thermal management, GPU-accessible memory and Windows ML integration for RTX Spark.

CPU performance

The P16's measured CPU results establish a foundation. Cinebench 2026 returned 557 in single-core and 7,117 in multicore. Geekbench 7.1.0 produced 2,699 single-core and 24,155 multicore.

Processor	Cinebench 2026 single-thread	Cinebench 2026 multicore
NVIDIA RTX Spark N1X, tested P16	557	7,117
Qualcomm Snapdragon X2 Elite Extreme X2E-96-100	668	7,412
Intel Core Ultra 9 290HX Plus	569	8,592
AMD Ryzen 9 9955HX	527	8,018
AMD Ryzen AI Max+ 395	467	7,268

Cinebench 2026 provides a consistent CPU rendering workload for comparing RTX Spark with competing Intel, AMD and Qualcomm processors.

N1X finishes close to the Core Ultra 9 290HX Plus and ahead of the Ryzen 9 9955HX in single-thread rendering. Qualcomm's X2 Elite Extreme is faster, scoring 668 against the P16's 557. In multicore rendering, the P16 finishes approximately 4% behind the X2 Elite Extreme and 2% behind the Ryzen AI Max+ 395 average. N1X is competitive with both, although laptop cooling and power limits can influence margins this small. Intel's 290HX Plus and AMD's 9955HX retain a clearer multicore advantage. The P16 trails their listed results by approximately 17% and 11%, respectively. For work dominated by CPU rendering, a powerful HX laptop remains a relevant alternative.

Game	Average frame rate
Cyberpunk 2077	138 FPS
DOOM: The Dark Ages	141 FPS
Gears of War: E-Day	135 FPS

AMD's Ryzen AI Max+ 395 is a close rival, combining an x86 CPU, Radeon graphics and shared memory. N1X delivers higher single-thread performance in Cinebench and finishes just behind it in multicore. CUDA support gives RTX Spark a different route to accelerating rendering and local AI, while AMD retains native compatibility with established x86 software. In Geekbench 7.1.0, N1X scored 2,699 single-core and 24,155 multicore.

Gaming performance

For most runs, we used DLSS Super Resolution and 3× frame generation at 1440p, with ray tracing enabled.

At these settings, RTX Spark delivers triple-digit displayed frame rates in demanding games at 1440p. DLSS upscaling and frame generation are central to that performance, alongside low quality settings in several titles.

Super Resolution reconstructs the output from a lower internal resolution, reducing the conventional rendering workload. Frame generation inserts additional displayed frames, while Ray Reconstruction replaces supported denoising paths. Each of these settings serves a different purpose.

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